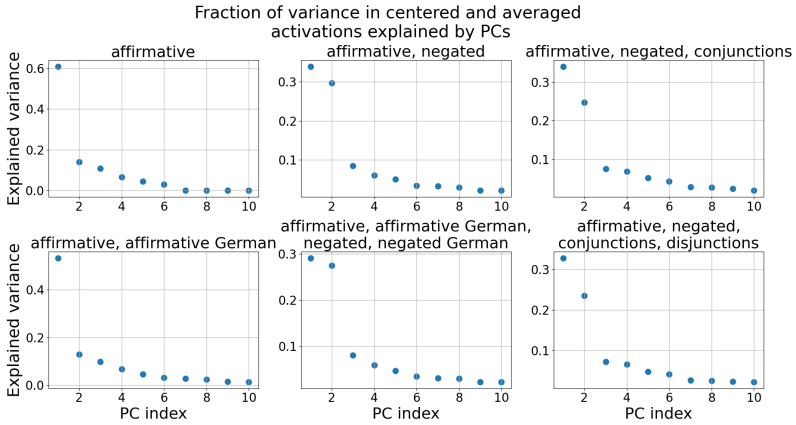

Figure 4: The fraction of variance in the centered and averaged activations $\tilde{\boldsymbol{\mu}}_i^+$, $\tilde{\boldsymbol{\mu}}_i^-$ explained by the Principal Components (PCs). Only the first 10 PCs are shown.

with the truth by averaging the activations:

$$\tilde{\boldsymbol{\mu}}_i^+ = \frac{2}{n_i} \sum_{j=1}^{n_i/2} \tilde{\mathbf{a}}_{ij}^+ \qquad \tilde{\boldsymbol{\mu}}_i^- = \frac{2}{n_i} \sum_{j=1}^{n_i/2} \tilde{\mathbf{a}}_{ij}^-, \tag{6}$$

where $\tilde{\mathbf{a}}_{ij}^+$ and $\tilde{\mathbf{a}}_{ij}^-$ are the centered activations corresponding to true and false statements, respectively. We then perform PCA on these preprocessed activations, including different statement types in the different plots. For each statement type, there are six topics and thus twelve centered and averaged activations $\tilde{\boldsymbol{\mu}}_i^\pm$ used for PCA.

Figure 4 illustrates our findings. When applying PCA to affirmative statements only (top left), the first PC explains approximately 60% of the variance in the centered and averaged activations, with subsequent PCs contributing significantly less, indicative of a one-dimensional affirmative truth direction. Including both affirmative and negated statements (top center) reveals a two-dimensional truth subspace, where the first two PCs account for more than 60% of the variance in the preprocessed activations. Note that in the raw, non-preprocessed activations they account only for $\approx 10\%$ of the variance. We verified that these two PCs indeed approximately correspond to $\mathbf{t}_G$ and $\mathbf{t}_P$ by computing the cosine similarities between the first PC and $\mathbf{t}_G$ and between the second PC and $\mathbf{t}_P$, measuring cosine similarities of 0.98 and 0.97, respectively. As shown in the other panels of Figure 4, adding logical conjunctions, disjunctions and statements translated to German does not increase the number of significant PCs beyond two, indicating that two principal components sufficiently capture the truth-related variance, suggesting only two truth dimensions.

4.2 Generalization of different truth directions

To further investigate the dimensionality of the truth subspace, we examine two aspects: (1) How well different truth directions $\mathbf{t}$ trained on progressively more statement types generalize; (2) Whether the activations of true and false statements remain linearly separable along some direction $\mathbf{t}$ after projecting out the 2D subspace spanned by $\mathbf{t}_G$ and $\mathbf{t}_P$ from the training activations. Figure 5 illustrates these aspects in the left and right panels, respectively. We compute each $\mathbf{t}$ using the supervised learning approach from Section 3, with all polarities p_i set to zero to learn a single truth direction.

In the left panel, we progressively include more statement types in the training data for $\mathbf{t}$: first affirmative, then negated, followed by logical conjunctions and disjunctions. We measure the separation of true and false activations along $\mathbf{t}$ via the AUROC. The right panel shows the separation along truth directions learned from activations $\bar{\mathbf{a}}_{ij}$ which have been projected onto the orthogonal complement of the 2D truth subspace:

$$\bar{\mathbf{a}}_{ij} = P^\perp(\mathbf{a}_{ij}), \tag{7}$$

where $P^\perp$ is the projection onto the orthogonal complement of $\text{Span}(\mathbf{t}_G, \mathbf{t}_P)$. We train all truth directions on 80% of the data, evaluating on the held-out 20% if the test and train sets are the same,